CONFIDENTIAL • APEX DYNAMICS INTERNAL OPERATIONS

Project Titan: Infrastructure & Incident Protocol

Standard Operating Procedures • Revision 3.2 • Effective Date: Q4 2026

1. Overview and Core Architecture

Project Titan represents Apex Dynamics' next-generation distributed transaction layer, designed to handle up to 85,000 requests per second with sub-12ms latency across the Asia-Pacific region. The primary compute workloads are orchestrated on isolated Kubernetes clusters running in AWS ap-south-1 (Mumbai) and ap-southeast-1 (Singapore).

2. Network Endpoints & Port Assignments

All microservices communicate via mutual TLS (mTLS) with strict ingress controls. Default internal port assignments are cataloged below:

Service Name	Internal Port	Protocol	Primary Maintainer
Titan Auth Gateway	8443	gRPC / mTLS	Elena Rostova
Ledger Synchronization Engine	9092	TCP Kafka Wire	David K. Chen
Telemetry & Audit Collector	50051	HTTP/2 gRPC	Amina Al-Mansoor
Administrative Dashboard	8501	HTTPS / WebSocket	DevOps SRE Team

3. Critical Severity Incident Escalation (#TITAN-CRIT-99)

In the event of database deadlock, split-brain consensus failure, or unauthorized egress detection, engineers on call must adhere strictly to the Tier-1 escalation timetable:

- **T + 0 min:** Automated alert fires into Slack channel `#ops-titan-alerts` and PagerDuty schedule Titan-Primary.
- **T + 5 min:** Incident Commander (Lead Architect: **Marcus Vance**) assumes communication control. Backup Commander is **Dr. Sarah Lin**.
- **T + 15 min:** All emergency issue tickets must be tagged with `#TITAN-CRIT-99` and assigned directly to the Security Incident Response Team (SIRT).
- **SLA Threshold:** Total Mean Time to Recovery (MTTR) must not exceed **28 minutes** to prevent breach of enterprise client contracts.

> **Mandatory Rollback Procedure:** If hotfix deployment verification exceeds 10 minutes without state convergence, the primary database replicas must be restored from snapshot `snap-titan-gold-latest` located in S3 bucket `s3://apex-titan-vault-prod`.